Does Displayed Reasoning Drive the Answer? A Diagnostic Benchmark for Vietnamese Educational Reasoning

Thanh Long Ly
11230561@st.neu.edu.vn
National Economics University
Hanoi, Vietnam

Thi Huyen Trang Duong
11230593@st.neu.edu.vn
National Economics University
Hanoi, Vietnam

Hoang Phuc Ngo
11230582@st.neu.edu.vn
National Economics University
Hanoi, Vietnam

Manh Toan Nguyen*
toannm36@fe.edu.vn
Swinburne Vietnam, FPT University
Vietnam

Abstract

Large language models increasingly assist Vietnamese learners, yet a fluent displayed rationale need not determine the final answer. We introduce a Vietnamese benchmark measuring behavioural dependence on displayed reasoning across computational, theoretical, and history/literature problem types. The candidate pool contains 72 matched triplets (216 items) across four education levels. For DeepSeek-R1-0528 under an unfinished-<think> continuation protocol, we compare continuation from clean, placebo-edited, and corrupted step-3 prefixes. Across 162 executed items (54 triplets; 9,720 generations), corrupted step-3 reasoning increases error propagation by 10.9 percentage points (95% bootstrap CI [7.3, 14.8], excluding zero), while placebo rewriting induces 0.0% propagation. In exploratory regression, higher answer-only accuracy is associated with lower propagation (pooled cluster-robust $\beta_q = -20.7$ pp per unit q_i, $p = 0.033$, uncorrected); slopes by problem type are not statistically distinguishable (omnibus $p = 0.075$). Auditing continuations across all 2,846 corrupted gold returns indicates that keyword-based silent override dominates (74.1% by keyword audit; $\approx$ 66%–74% after calibration), with explicit repair $\approx$ 26%–34%. Failing the factual-prerequisite probes was not associated with more propagation (10.5% vs. 13.4%), although items needing external facts propagated more often than self-contained ones (descriptive, confounded with problem type). Data and code are available at https://github.com/tlong-ds/benchmark-vnedu-reasoning.

CCS Concepts

• **Computing methodologies** → **Natural language processing**;
• **Applied computing** → **Education**.

*Corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
ICIIT 2027, Ho Chi Minh City, Vietnam
© 2027 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-x-xxxx-xxxx-x/YY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

Keywords

chain-of-thought, reasoning faithfulness, AI tutoring, Vietnamese benchmark, large language models

ACM Reference Format:
Thanh Long Ly, Thi Huyen Trang Duong, Hoang Phuc Ngo, and Manh Toan Nguyen. 2027. Does Displayed Reasoning Drive the Answer? A Diagnostic Benchmark for Vietnamese Educational Reasoning. In *Proceedings of the 2027 12th International Conference on Intelligent Information Technology (ICIIT 2027), March 4–7, 2027, Ho Chi Minh City, Vietnam.* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

Large language models (LLMs) increasingly serve as conversational study aids whose educational value depends on visible explanations: learners inspect intermediate steps rather than passively accepting answers, aligning with policy guidance on AI in higher education emphasizing transparency [16]. However, inspecting displayed reasoning is informative only if that reasoning actually drives the final answer.

That assumption can fail. Models can omit influential cues from their explanations [15], keep the same answer despite severe perturbations to intermediate steps [8], and produce unfaithful chains on natural inputs [1]. We study *behavioural dependence on displayed reasoning*: whether editing one intermediate step changes the answer the model goes on to give.

Vietnamese adds its own difficulties: reasoning can collapse into English [9], misalign across languages [11], and stay inconsistent across languages for the same problem [17]. Comparing problem types is also confounded, because mathematics and history differ at once in deductive structure, factual recall, and how directly the answer can be recalled. We separate these factors with matched triplets, prerequisite knowledge probes, an answer-only baseline without native reasoning tokens, and blinded human validation.

Contributions. (1) We construct 72 *matched triplets* (216 Vietnamese items) of computational, theoretical, and history/literature problems at four education levels (primary to university); the items of a triplet share an inference schema, chain shape, and answer structure. (2) We introduce a three-arm continuation intervention (clean, placebo, corrupted) that separates error propagation from the effect of rewording. (3) We add two controls: standalone *knowledge probes* for prerequisite facts, recorded rather than used to filter items (90/118 probed items passed; 134/162 overall), and an

answer-only baseline without native reasoning tokens (q_i, $k_{\mathrm{AO}} = 10$, 2,160 draws). (4) For DeepSeek-R1-0528 on 54 executed triplets (162 items; 9,720 generations), we estimate corruption dependence and its association with answer-only accuracy (negative pooled association; slopes by problem type are not statistically distinguishable), and describe how continuations respond to a corrupted step: adopting it without overt doubt, silently overriding it, or explicitly repairing it. Data and code are available at https://github.com/tlong-ds/benchmark-vnedu-reasoning.

2 Related Work

Two intervention paradigms evaluate chain-of-thought fidelity: bias and hint injection alter external cues to test disclosure [15], while trace perturbation alters intermediate reasoning steps to measure prediction shifts [4, 8]. Recent work evaluates naturally occurring unfaithful chains [1], benchmark-level faithfulness audits [13], and LLM judges of causal/coverage faithfulness [10]. While multilingual reasoning has been studied globally [17], Southeast Asian languages often experience cross-lingual collapse to English [9] and reasoning misalignment [11]. Vietnamese evaluation resources include VNHSGE [5], VMLU [3], and specialized benchmarks [6, 14]. Educational evaluations focus primarily on dialogue quality [2, 7] or truncated audits [12]. To our knowledge, this is the first Vietnamese benchmark that uses matched triplets to compare reasoning dependence across problem types.

3 Benchmark Design

3.1 Matched Triplets

The design unit is a *triplet*: one computational, one theoretical and one history/literature item that share (i) an abstract inference schema, (ii) a four-step reasoning shape, (iii) the edited step, always step 3, and (iv) matched answer options, with gold and propagation positions balanced. The candidate pool contains 72 triplets (216 items), 18 at each of four education levels (primary to university). Triplets instantiate six inference schemas: S1 (sequential dependence), S2 (necessary treated as sufficient), S3 (elimination), S4 (causal attribution), S5 (comparison differing in one variable), and S6 (simultaneous constraints).

3.2 Knowledge Probes and Quality Constraints

Facts an item needs but does not state are asked as standalone probes before the intervention. Probe accuracy is recorded as a covariate, not used to exclude items (Section 6.4); items with no external premises score 1.0 by construction. Items meet five design constraints: (1) *propagation uniqueness* (exactly one option follows from the corrupted inference); (2) *no answer in step 3* (step 3 states an intermediate result, never an option letter or final answer, so the answer cannot be copied; every item was checked before execution); (3) *option balance* (lengths within ±25%, positions balanced); (4) *no world-knowledge refutation* (the propagation answer must not be trivially refutable from world knowledge alone; recall shortcuts to gold are measured by answer-only accuracy q_i (Section 4), not excluded); and (5) *deduplication*.

3.3 Human Validation

Two independent raters judged blinded prefixes (steps 1–3, without step 4 or model continuations; the clean, placebo and corrupted arms are defined in Section 4): detectability and answer relevance for corrupted steps, meaning preservation for placebo rewrites, and salted clean catch trials. Detectability was rated on an ordinal 0–5 scale, so inter-rater agreement for severity was assessed with quadratic-weighted Cohen's κ,

$$\kappa_w = 1 - \frac{\sum_{i,j} w_{ij} O_{ij}}{\sum_{i,j} w_{ij} E_{ij}}, \qquad w_{ij} = \frac{(i-j)^2}{25}, \tag{1}$$

where O_{ij} is the share of prefixes scored i by Rater 1 and j by Rater 2, and E_{ij} the share expected from the raters' marginal score distributions. A value of $\kappa_w = 1$ is perfect agreement and 0 is chance level; the raters reached $\kappa_w = 0.28$. The human-screen and detectability results are in Sections 6.1 and 6.4.

4 Intervention and Measures

All prefixes derive from a four-step *base chain*: the focal model (DeepSeek-R1-0528) solves each item at temperature 0, and its reasoning is condensed into four steps and edited into a uniform step format. Each item has three prefixes of equal structure ending at step 3: (1) **clean**, the base chain's step 3; (2) **placebo**, a meaning-preserving rewrite that controls for rewording; and (3) **corrupted**, an invalid inference on which the answer depends. In a chemistry item, for example, step 3 reads "$n_{SO_2} = 0.1$ mol" (clean), "Số mol SO_2 là 0.1 mol" (placebo), or "$n_{SO_2} = 0.2$ mol" (corrupted). We hypothesized that dependence would be highest in computation and lowest in history (H1: Δ_{CDE} for computational > theoretical > history/literature items, with Δ_{CDE} defined in Eq. 3 below): an intermediate error in a calculation forces a different answer, whereas history items can often be answered from recall.

The primary binary outcome indicates selection of the propagation answer,

$$Y = \mathbf{1}[\text{final answer} = \text{propagation answer}]. \tag{2}$$

For problem type t, the primary *Corruption Dependence Effect* is

$$\Delta_{\mathrm{CDE}}(t) = \Pr(Y = 1 \mid \text{corrupted}, t) - \Pr(Y = 1 \mid \text{placebo}, t). \tag{3}$$

The clean arm provides a descriptive manipulation check,

$$\Delta_{\mathrm{edit}}(t) = \Pr(Y = 1 \mid \text{placebo}, t) - \Pr(Y = 1 \mid \text{clean}, t). \tag{4}$$

To measure answer-only accuracy, that is, accuracy without displayed reasoning, we suppress native reasoning tokens with a prefilled empty thinking block (<think>\n</think>\nĐáp án:) over $k_{\mathrm{AO}} = 10$ cyclic option rotations, giving

$$q_i = \Pr(\text{gold} \mid \text{answer-only}, i). \tag{5}$$

As an exploratory measure, corrupted-arm continuations are also classified by behaviour (explicit repair, silent override, premise adoption without overt doubt).

5 Experimental Setup

Interventions evaluate DeepSeek-R1-0528 (served by DeepInfra via OpenRouter, October 2026). Prefixes are injected into an unfinished assistant <think> block with high reasoning effort (mapped to maximum thinking budget by DeepInfra) and provider fallbacks disabled.

Sampling uses temperature 0.7, top-p = 1, max 8,192 tokens, and k = 20 draws per arm (9,720 generations across 162 items in 54 triplets). Per-draw deterministic seeds allow auditing, and a per-draw check confirms that the provider continued the open <think> block rather than closing or restarting it. Placebo propagation is zero, which makes logistic models degenerate (complete separation), so we report non-parametric whole-triplet bootstrap intervals (10,000 resamples; seed 20261013) for $\Delta_{\text{CDE}}(t)$ and within-triplet contrasts. Regressions model item-level corrupted-arm propagation (n = 162 items) on q_i and problem-type indicators, with triplet cluster-robust standard errors (G = 54).

6 Results

6.1 Benchmark Coverage and Quality Checks

The candidate pool comprises 216 items (72 triplets) across four educational levels. Eighteen candidate triplets were withheld because a base chain missed the gold answer or failed generation, format, or answer-parsing checks, leaving 162 executed items in 54 complete triplets (54 per problem type; 42 each at primary, secondary, high school; 36 university). Each item has 20 records per arm: 9,720 generations (9,707 parseable answers, 13 unparseable responses retained in denominators). Protocol diagnostics flag 270 records lacking native reasoning content (66 clean, 123 placebo, 81 corrupted); 140 items have no flagged record in any arm ("protocol-complete"; 38 triplets have all three items complete). A language audit of the 9,450 continuations with reasoning content finds all of them predominantly Vietnamese (none English; 4 contain a few isolated English words). Across the six abstract inference schemas (n = 8–10 triplets each), descriptive dependence ranges from 3.7% in S4 to 17.5% in S1 and S3 (Table 2). Blinded paired human ratings cover all 162 corrupted prefixes and 162 placebo comparisons (3 clean catch false alarms in 58 judgments, 5.2%). Both raters confirmed meaning preservation for 122/162 placebos (19 rejected by both, 21 split). The primary effect remains on the subset passing all paired human screens (Δ_{CDE} = 12.1 pp, 95% CI [8.2, 16.6]), indicating that disputed placebo rewrites do not explain the result.

6.2 Primary Corruption-Dependence Estimates

Table 1: Propagation rates, corrupted-versus-placebo effects (Δ_{CDE}), and overall non-gold rates on the executed items (54 triplets, 162 items, 9,720 draws). CIs are percentage points from whole-triplet resampling (10,000 replicates).

Problem type	Placebo	Corrupted	Δ_{CDE} (95% CI)	Non-gold (Corr.)
Computational	0.0%	8.8%	8.8 [3.4, 15.2]	9.4%
Theoretical	0.0%	10.4%	10.4 [5.0, 16.6]	13.4%
History/literature	0.0%	13.4%	13.4 [6.8, 20.8]	13.6%
All items	0.0%	10.9%	10.9 [7.3, 14.8]	12.2%

Table 1 shows that the pooled corrupted-versus-placebo effect is 10.9 percentage points (95% CI: [7.3, 14.8]; 352/3,240 corrupted vs. 0/3,240 placebo, 5/3,240 clean). The placebo–clean contrast Δ_{edit} (Eq. 4) is −0.15 pp (95% CI: [−0.5, 0.0]), so rewording alone has almost no effect (rule-of-three bound ≈ 0.09%). Non-gold answers occur in 0.4% clean, 0.1% placebo, and 12.2% corrupted draws.

Table 2: Δ_{CDE} (percentage points) by abstract inference schema. Placebo propagation is zero in every schema, so Δ_{CDE} equals the corrupted propagation rate.

Schema	Triplets	Propagating draws	Δ_{CDE} (95% CI)
S1	8	84/480	17.5 [6.7, 32.1]
S2	9	37/540	6.9 [1.9, 12.6]
S3	8	84/480	17.5 [7.3, 28.5]
S4	10	22/600	3.7 [0.0, 10.2]
S5	10	46/600	7.7 [1.7, 15.0]
S6	9	79/540	14.6 [6.3, 24.4]

Propagation is concentrated in few items: 116 of 162 items (71.6%) exhibit zero propagation across all 20 draws, 45 items (27.8%) show partial propagation (1–19 draws), and exactly 1 item (0.6%) propagates in all 20 draws (within-item intraclass correlation ICC_{item} = 0.58, design effect DEFF = 1 + 19 ICC_{item} = 12.0, within-triplet $\text{ICC}_{\text{triplet}}$ = 0.21). Between computational and history/literature problems, the unadjusted paired within-triplet contrast is −4.6 pp (95% CI: [−14.3, 5.0]); theoretical vs. history is −3.1 pp (95% CI: [−11.8, 5.1]); contrasts use unrounded rates. Point estimates reverse the hypothesized H1 ordering (history 13.4% > theoretical 10.4% > computation 8.8%), though both contrast intervals span zero. The effect holds on stricter subsets (Table 3): its interval excludes zero on the 140 protocol-complete items, on the 117 items passing all paired human screens, and on the 88 items that are human-screened, protocol-complete and pass the knowledge probes.

Table 3: Δ_{CDE} (percentage points) on stricter subsets, with whole-triplet bootstrap intervals. "Complete" means protocol-complete. "Human-screened" requires both raters to judge the corruption detectable and answer-relevant and to confirm placebo meaning preservation.

Subset	Items	Δ_{CDE} (95% CI)
All executed items	162	10.9 [7.3, 14.8]
Protocol-complete	140	10.4 [6.5, 14.9]
Human-screened	117	12.1 [8.2, 16.6]
Human-screened, complete, probes passed	88	13.4 [8.3, 19.0]

6.3 Answer-Only Accuracy and Problem Type

Table 4: Answer-only accuracy q_i and corruption dependence Δ_{CDE} (percentage points). β_q is the change in item-level corrupted propagation (pp) per unit q_i with triplet cluster-robust SEs (G = 54): the pooled slope is from a model with problem-type indicators, type slopes are within-type. Strata are descriptive; p tests the slope.

		Slope on q_i		Δ_{CDE} by q_i stratum (n)		
Problem type	q_i	β_q (SE)	p	≥.85	.60–.85	<.60
Computational	69.8%	−29.4 (14.9)	0.053	3.7 (23)	7.9 (17)	18.2 (14)
Theoretical	76.1%	−30.2 (18.4)	0.107	8.4 (22)	6.8 (25)	29.3 (7)
History/literature	81.7%	+8.3 (10.1)	0.413	12.3 (28)	17.6 (21)	2.0 (5)
All items	75.9%	−20.7 (9.5)	0.033	8.4 (73)	10.7 (63)	18.1 (26)

Answer-only accuracy q_i uses an empty thinking block ($k_{AO} = 10$ cyclic option rotations on all 216 items; 2,160 draws with no native reasoning tokens, confirmed via usage metadata). Mean accuracy on executed items is 75.9% (vs. 70.9% on the 54 withheld items), and computational items have the lowest accuracy (Table 4).

In exploratory regression, answer-only accuracy is negatively associated with corruption dependence overall ($\beta_q = -20.72$ pp per unit q_i, $p = 0.033$). The omnibus $q_i \times$ type interaction is not significant ($\chi^2(2) = 5.19$, $p = 0.075$; computational interaction uncorrected $p = 0.033$, Bonferroni $p = 0.066$; theoretical uncorrected $p = 0.119$, Bonferroni $p = 0.238$), so the slopes by problem type in Table 4 are not statistically distinguishable. Adding the answer-only rate of choosing the propagation option (mean 5.7%) as a covariate changes little ($\beta = 1.46$, $p = 0.933$).

Stratifying by q_i is descriptive and not a test of differences between problem types. On computational and theoretical items, high-accuracy items propagate rarely and low-accuracy items more often, while history/literature shows no such gradient: even high-accuracy history items propagate at 12.3%, and the low-accuracy history stratum has only five items. Adjusting for q_i shifts both contrasts with history further negative (computational vs. history: −7.09 pp, 95% CI [−16.3, 2.1]; theoretical vs. history: −4.21 pp, 95% CI [−12.9, 4.4]; both span zero), so H1 is not recovered.

6.4 Knowledge Probes and Rater Detectability

Knowledge probes (Section 3.2) were run for every item that has external premises. Forty-four computational items need no external facts and pass the probes by construction. Of the 118 items that do, the model recalls every probed fact for 90 items (76.3%: 9/10 computation, 43/54 theoretical, 38/54 history/literature; 134/162 overall; 15 empty responses count as failures). Failing items show $\Delta_{CDE} = 10.5\%$ ($n = 28$) vs. 13.4% for passing ones ($n = 90$). Items requiring external factual recall propagate at about twice the rate of self-contained items (12.8% vs. 5.8%); within computational items alone, 10 external-load items propagate at 22.0% (95% CI [2.5, 46.0]) vs. 5.8% for 44 zero-load items (95% CI [1.6, 11.2]). Since zero-load items are all computational, this is descriptive; adjusting for q_i and problem type (identified within computational items) gives $\beta_{ext} = +15.0$ pp (SE = 10.2, $p = 0.147$, spanning zero).

Detectability (0–5) averages 3.73 (SD 1.06) with a similar spread in every problem type, and shows negligible correlation with Δ_{CDE} ($r = -0.096$). This check is limited by modest rater severity agreement ($\kappa_w = 0.28$, Eq. 1). Quadratic weights penalise large score gaps more than small ones. As Table 5 shows, 78% of score pairs differ by at most one point, but Rater 1 scores higher on average (4.06 vs. 3.41) and the raters correlate only weakly ($r = 0.35$). Detectability is therefore a coarse covariate, and its null association with Δ_{CDE} is weak evidence, not proof of no relation.

6.5 Behavioural Continuation Breakdown

Table 6 splits the corrupted draws. Across the 2,846 gold returns, 74.1% (2,109 draws) are classified by the lexical audit as *silent override*: no error-repair marker. On 76 human-rated continuations from a mixed-type calibration sample, the corpus keyword classifier has precision/recall 0.65/0.50 (Rater 1) and 0.47/0.47 (Rater 2); adjusting the flag rate by precision/recall gives explicit repair ≈ 26%–34%, so the raw 25.9% sits at the low end of that range and silent override is ≈ 66%–74% (still the majority). Conditioning on gold returns, silent override dominates on high-q_i items ($q_i \geq 0.85$, 77.6%, $n = 1{,}034/1{,}333$), whereas apparent explicit repair (descriptive) rises on low-q_i items ($q_i < 0.60$, 31.1%, $n = 131/421$) and low-q_i quantitative items (35.0%, $n = 113/323$). However, because verification keywords (*tính lại*, *xét lại*) concentrate in arithmetic tasks, this apparent quantitative rise may partly reflect recalculation false alarms rather than true behavioral correction. The conflict lexicon used for the propagation returns is narrow and unvalidated.

Table 5: Agreement between the two raters on corruption detectability. Counts are prefixes.

Detectability (0–5), 162 prefixes	Value
Identical scores	62
Differ by one point	65
Differ by two or more points	35
Rater 1 scores higher / lower	82 / 18
Mean score, Rater 1 / Rater 2	4.06 / 3.41
Pearson r	0.35
Quadratic-weighted κ_w	0.282

Table 6: Outcomes of the 3,240 corrupted-arm draws. Repair and doubt are rule-based lexical flags (see text for their precision and recall).

Corrupted-arm outcome	Draws	Share
Gold answer	2,846	87.8%
silent override (no repair marker)	2,109	74.1% of gold
explicit repair marker	737	25.9% of gold
Propagation answer	352	10.9%
no overt doubt	343	97.4% of propagation
doubt expressed	9	2.6% of propagation
Other option	29	0.9%
Unparseable	13	0.4%
Total	3,240	100%

7 Discussion

The benchmark separates prerequisite knowledge (probes), answer-only accuracy (q_i), and dependence on displayed reasoning (Δ_{CDE}). Probe failures (90/118 pass) are not associated with more propagation, so we find no evidence that missing prerequisite knowledge drives it, although items needing external facts propagate more often than self-contained ones (descriptive). Facing corrupted premises, the model either adopts them without overt doubt or returns gold via keyword-based silent override (74.1% of gold draws by keyword audit; ≈ 66%–74% calibrated) or explicit repair (≈ 26%–34%).

H1 (Δ_{CDE} highest for computational, lowest for history/literature items) was not supported: point estimates reversed it (history 13.4% > theoretical 10.4% > computation 8.8%; CIs span zero), and adjusting for q_i did not recover it. In exploratory regression, the pooled association between answer-only accuracy and propagation is negative ($\beta_q = -20.72$ pp, $p = 0.033$, uncorrected), while slopes by problem type are not statistically distinguishable (omnibus $p = 0.075$). Problem type itself therefore did not explain dependence as expected: answer-only accuracy was associated with dependence

overall, but its slope did not differ significantly by problem type. Even high-accuracy history items ($q_i \geq 0.85$; $n = 28$, descriptive) propagate at 12.3%, which is compatible with deference to displayed premises even when recall would suffice but does not establish it.

For tutoring, propagation (10.9% of corrupted draws) rarely voices doubt (no overt doubt in 97.4% of propagating draws by a narrow, unvalidated conflict lexicon), a potential pedagogical hazard: a learner sees a wrong step carried through without warning. Conversely, keyword-based silent override, the majority of gold returns (descriptive; confounded by problem-type mix), suggests that a correct answer can follow an uncorrected wrong step, so a displayed rationale cannot be presumed to be what produced the answer.

8 Limitations

While cross-lingual collapse motivates the Vietnamese setting, our evaluation operates under controlled prefilling without a parallel English control; unprompted reasoning is not evaluated. Our evaluation is a diagnostic case study on DeepSeek-R1-0528; deterministic seeds enable auditing but do not guarantee bit-identical outputs across infrastructure updates. We intervene within unfinished <think> traces with condensed, edited prefixes rather than verbatim model reasoning; visible dialogue turns may exhibit different dynamics. Corruptions were fairly salient (detectability 3.73/5), so subtler errors may propagate more. Withholding 18 candidate triplets (mainly failing unperturbed base-chain accuracy or format constraints) introduces range restriction ($q_i = 75.9\%$ vs. 70.9% in discarded items), yielding a small low-accuracy stratum and potential slope attenuation at low q_i. The two raters worked separately, without access to model continuations or outcome statistics, and finished before the experimental results were aggregated; their modest agreement on detectability ($\kappa_w = 0.28$) limits the precision of severity-based analyses. The lexical audit exhibits substantial false alarms. Open-world history items confound deduction with parametric recall, and exploratory comparisons are unadjusted for multiplicity.

9 Ethical Considerations

The benchmark evaluates AI systems without student subjects or personal data, supporting pedagogical transparency. Prompts, metadata and code are released in the public repository.

10 Conclusion

We evaluate dependence on displayed reasoning in Vietnamese educational problems across 162 executed items. Corrupted step-3 reasoning increases error propagation by 10.9 pp (95% bootstrap CI [7.3, 14.8], excluding zero). Answer-only accuracy is negatively associated with corruption dependence (pooled $\beta_q = -20.72$ pp per unit q_i, $p = 0.033$, uncorrected); slopes by problem type are not statistically distinguishable (omnibus $p = 0.075$). For this model and protocol, editing step 3 changed the final answer in a minority of draws: 10.9% of corrupted draws propagate the injected error, while 28.4% of items propagate it in at least one of 20 draws.

References

[1] Iván Arcuschin, Jett Janiak, Robert Krzyzanowski, Senthooran Rajamanoharan, Neel Nanda, and Arthur Conmy. 2026. Chain-of-Thought Reasoning In The Wild Is Not Always Faithful. In *Proceedings of ICML 2026*. 3354–3405. arXiv:2503.08679.

[2] Parth Bhalerao, Jeromy Chang, David Chou, and Oana Ignat. 2026. GRADE: Generalizable Reasoning-Aware Dialogue Evaluation for AI Tutors. *arXiv:2605.27866* (2026).

[3] Cuc Thi Bui, Nguyen Truong Son, Truong Van Trang, Lam Viet Phung, Pham Nhut Huy, Hoang Anh Le, Quoc Huu Van, Phong Nguyen-Thuan Do, Van Le Tran Truc, Duc Thanh Chau, et al. 2025. VMLU Benchmarks: A comprehensive benchmark toolkit for Vietnamese LLMs. In *Proceedings of ACL*. 11495–11515.

[4] Yanda Chen, Joe Benton, Ansh Radhakrishnan, Jonathan Uesato, Carson Denison, John Schulman, Arushi Somani, Peter Hase, Misha Wagner, Fabien Roger, et al. 2025. Reasoning models don't always say what they think. *arXiv:2505.05410* (2025).

[5] Xuan-Quy Dao et al. 2023. VNHSGE: Vietnamese High School Graduation Examination Dataset for Large Language Models. *arXiv:2305.12199* (2023).

[6] Nguyen Tien Dong, Minh-Anh Nguyen, Thanh Dat Hoang, Nguyen Tuan Ngoc, Dao Xuan Quang Minh, Phan Phi Hai, Nguyen Thi Ngoc Anh, Dang Van Tu, and Binh Vu. 2025. VLegal-Bench: Cognitively Grounded Benchmark for Vietnamese Legal Reasoning of Large Language Models. *arXiv:2512.14554* (2025).

[7] Ekaterina Kochmar, Kaushal Kumar Maurya, Kseniia Petukhova, KV Aditya Srivatsa, Anaïs Tack, and Justin Vasselli. 2025. Findings of the BEA 2025 shared task on pedagogical ability assessment of AI-powered tutors. In *Proceedings of BEA Workshop*. 1011–1033.

[8] Tamera Lanham, Anna Chen, Ansh Radhakrishnan, et al. 2023. Measuring Faithfulness in Chain-of-Thought Reasoning. *arXiv:2307.13702* (2023).

[9] Sean Gip Lim, William Chandra Tjhi, and Hai Leong Chieu. 2026. Native Multilingual Chain-of-Thought Reasoning in Low-Resource Southeast Asian Languages. *arXiv:2608.00533* (2026).

[10] Avni Mittal and Rauno Arike. 2026. C2-Faith: Benchmarking LLM Judges for Causal and Coverage Faithfulness in Chain-of-Thought Reasoning. In *Proceedings of GEM Workshop*. 678–691.

[11] Anaelia Ovalle, Candace Ross, Sebastian Ruder, Adina Williams, Karen Ullrich, Mark Ibrahim, and Levent Sagun. 2025. Beg to differ: Understanding reasoning-answer misalignment across languages. *arXiv:2512.22712* (2025).

[12] Bonan Shen, Dingyan Shang, Youting Wang, and Tao Ning. 2026. Detecting Answer-Driven Reasoning in LLM-Based Educational Tutors via Truncated Chain-of-Thought Auditing. *arXiv:2607.04572* (2026).

[13] Xu Shen, Song Wang, Zhen Tan, Laura Yao, Xinyu Zhao, Kaidi Xu, Xin Wang, and Tianlong Chen. 2026. Faithcot-bench: Benchmarking instance-level faithfulness of chain-of-thought reasoning. In *Proceedings of ICLR 2026*.

[14] Bui The Trung, Do Minh Duc, Nguyen Van Vinh, and Bui Nguyen Quoc Trinh. 2026. Bridging the Reasoning Gap in Vietnamese with Small Language Models via Test-Time Scaling. *arXiv:2604.17794* (2026).

[15] Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. 2023. Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting. In *NeurIPS*, Vol. 36. 74952–74965.

[16] Vietnam Ministry of Education and Training. 2026. Circular No. 49/2026/TT-BGDDT on AI in higher and vocational education. Vietnam Government Portal, https://chinhphu.vn.

[17] Raoyuan Zhao, Yihong Liu, Hinrich Schütze, and Michael A. Hedderich. 2026. A Comprehensive Evaluation of Multilingual Chain-of-Thought Reasoning: Performance, Consistency, and Faithfulness Across Languages. In *Findings of EACL*. 5223–5247.